

MUYIWA SUNDAY ALE

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[LinkedIn](#) | [GitHub](#)

PROFILE

Computer Science graduate with practical experience in software development, machine learning, computer vision, and data-driven modelling. Experienced in developing machine learning solutions from data preparation and model training through evaluation and application deployment. My project work includes real-time sign language recognition, medical image classification, predictive modelling, and web-based applications. I also have experience teaching Computer Science and leading community projects, giving me a combination of technical, communication, leadership, and problem-solving skills. I am building my career as a Software and Machine Learning Engineer, with an interest in developing practical intelligent systems that address real-world problems.

RESEARCH INTEREST

- Machine Learning & Deep Learning
- Computer Vision & Pattern Recognition
- AI for Healthcare Applications
- Real-Time Intelligent Systems
- Software Engineering
- Human-Computer Interaction

EDUCATION

Bachelor of Science in Computer Science

Federal University Lokoja December 2019 - September 2024

Second Class Upper Division (CGPA: 4.18/5.00)

Address: Lokoja, Kogi State, Nigeria

Thesis: Development of a Deep Learning-Based Static Sign Language Recognition System

RESEARCH, PROJECT & DIGITAL WORK

Deep Learning-Based Static Sign Language Recognition System

Undergraduate Thesis | Computer Vision | Deep Learning

September 2023 - September 2024

GitHub repository: [Link](#)

- Designed and trained a Convolutional Neural Network (CNN) for recognising static hand gestures using a custom image dataset.
- Implemented hand detection and real-time gesture tracking using OpenCV and computer vision techniques.
- Developed an interactive interface for demonstrating real-time gesture classification through webcam input.
- Applied image preprocessing, model evaluation, and parameter tuning to improve recognition performance.
- Explored the application of computer vision and deep learning to assistive communication for deaf and hard-of-hearing users.

CNN-Based Eye Disease Classification System

Independent Machine Learning Project | Computer Vision | Medical AI

March 2025 - June 2025

GitHub repository: [Link](#)

- Developed a CNN-based image classification model for identifying cataract, glaucoma, diabetic retinopathy, and normal retinal fundus images.
- Prepared and augmented image data to support model training and improve generalisation.
- Achieved 88% test accuracy during model evaluation.

- Deployed the trained model through a Flask-based web application with confidence-based predictions.
- Explored the use of computer vision and deep learning for medical image classification and early screening support.

Predictive Modelling for Drug Abuse Impact on Academic Performance

Independent Machine Learning Project | Predictive Analytics | Classification

January 2025 - March 2025

GitHub repository: [Link](#)

- Developed a machine learning solution to investigate the relationship between drug abuse indicators and academic performance.
- Collected and prepared data covering academic, behavioural, and socioeconomic factors for predictive modelling.
- Applied the CRISP-DM methodology to structure the project from data understanding through model evaluation.
- Compared Random Forest, Logistic Regression, Support Vector Machine, and K-Nearest Neighbours classification models.
- Achieved 92% accuracy with Logistic Regression during model evaluation.
- Developed a web-based interface for interactive data input and prediction.

Environmental CDS Giade Website

Community Digital Project | Web Development

July 2026

Website: [Link](#)

- Designed and deployed the official website for the Environmental & Climate Action CDS Group in Giade to centralise information about its projects, activities, and initiatives.
- Structured and organised project information into a public-facing digital platform for easier access and communication.
- Collaborated with CDS team members to gather, review, and organise content for publication.
- Maintained and updated the website to keep project information current and accessible.
- Applied HTML, CSS, JavaScript, and web development practices to deliver a practical digital solution for a community organisation

PROFESSIONAL EXPERIENCE

Computer Science Teacher

National Youth Service Corps (NYSC) | Government Day Secondary School, Giade, Bauchi State

October 2025 - August 2026

- Taught Computer Science to senior secondary students, covering programming fundamentals and core computational concepts.
- Designed lesson plans, practical exercises, and assessments aligned with curriculum requirements.
- Guided students in problem-solving, logical reasoning, and analytical thinking.
- Used practical activities to make technical concepts easier for students to understand and apply.

Private Academic Tutor

August 2023 - December 2023

- Delivered personalised Mathematics and English instruction based on individual learning needs.
- Adapted teaching approaches and lesson plans to support students' academic development.
- Provided guidance and feedback to help students improve their understanding of core concepts.

Classroom Teacher

Destiny Academy, Owo, Ondo State

September 2017 - July 2019

- Taught foundational subjects and supported students' development in analytical and problem-solving skills.
- Assisted with classroom activities, lesson preparation, and student engagement.

LEADERSHIP & VOLUNTEERING

President, Environmental & Climate Action CDS Group

Giade, Bauchi State

February 2026 - August 2026

- Coordinated team members in planning and executing community development projects and activities.
- Led the development and launch of the group's official website.
- Managed project information, documentation, and reporting.
- Organised environmental awareness activities and community programmes.
- Coordinated team communication, responsibilities, and project requirements.
- Supported initiatives including environmental clean-up activities and tree-planting programmes.

TECHNICAL SKILLS

Programming & Software Development:

Python, Java, JavaScript, HTML, CSS, Git, GitHub

Machine Learning & Data Science:

Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib

Computer Vision:

OpenCV, MediaPipe, cvzone

Data & Information Management:

Data Collection, Data Cleaning, Data Processing, Data Analysis, Data Organisation, Data Validation, Information Management, Technical Documentation, Reporting

Development & Data Tools:

Visual Studio Code, Jupyter Notebook, Anaconda

Microsoft Office:

Microsoft Excel, Word, PowerPoint, Access

IT & Technical Support:

Computer Systems, Software Troubleshooting, Software Installation & Configuration, ICT Operations, User Support, Technical Problem-Solving

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

AWS AI Practitioner Challenge

Amazon Web Services (AWS)

June 2026

Career Essentials in Generative AI

Microsoft & LinkedIn

August 2025

Certificate: [Link](#)

COMMUNITY

Member: Google Developer Group (GDG)

January 2021 - Present.